

# Phase 1: Requirement Analysis & Planning

**Goal:** turn the idea into a concrete, buildable scope before any configuration starts.

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## Business Objectives

- Eliminate manual/counter-based ticketing
- Ensure consistent, error-free fare pricing
- Provide an instant QR-based digital ticket
- Give operations visibility into usage & revenue

## Functional Scope

1. Capture source, destination, journey date, passenger count
2. Auto-calculate fare based on route
3. Auto-create ticket record with unique ticket number
4. Auto-generate QR code and attach to ticket
5. Email confirmation with ticket + QR
6. Dashboards for bookings, revenue, route trends

## Out of Scope

- Online payment processing
- Offline QR validation
- Dynamic/peak-hour pricing

## Stakeholder Mapping

| Stakeholder | Need                      | Touchpoint   |
|-------------|---------------------------|--------------|
| Passenger   | Fast self-service booking | Booking form |

| Stakeholder     | Need                        | Touchpoint          |
|-----------------|-----------------------------|---------------------|
| Operations Team | Demand & revenue visibility | Dashboard           |
| Admin           | Stable, secure config       | Tables, Rules, ACLs |
| Support Staff   | View/resolve issues         | Ticket list view    |

## Assumptions

- Single developer instance used for build & test
- Fare mapping is static per route
- All users authenticate via existing ServiceNow login

## Risks & Mitigations

| Risk                        | Mitigation                                                      |
|-----------------------------|-----------------------------------------------------------------|
| Scope creep (e.g. payments) | Explicit out-of-scope list agreed upfront                       |
| External QR API downtime    | Documented as a known limitation, retry logic planned           |
| Delays in later phases      | Roadmap sequenced so no phase blocks on an undefined dependency |

## Success Criteria

- All 6 functional scope items implemented
- All test cases in Phase 4 pass
- Demo in Phase 5 runs without manual intervention

## Execution Roadmap

1. Phase 2 — Backend Development & Configurations
2. Phase 3 — UI/UX Development & Customization
3. Phase 4 — Data Migration, Testing & Security
4. Phase 5 — Deployment, Documentation & Final Presentation

**Deliverable:** signed-off scope (objectives, functional scope, stakeholders, risks, and roadmap).